Household Financial Fragility Index (HFFI): A Quantitative Framework for Investment Strategy and Recommendation Systems

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***Abstract*—The process of financial decision-making at the household level may not be structured and may not have a quantitative base. In general, most individuals make financial decisions based on their judgment, market trends, and financial advice without assessing their financial resilience. This paper proposes a Household Financial Fragility Index (HFFI), a quantitative framework that measures financial vulnerability by integrating liquidity, debt, expense rigidity, portfolio risk, and macroeconomic sensitivity into a unified score.**

The proposed system enhances the above index by integrating it with a quantitative investment strategy engine and a rule-based recommendation system. In this way, the system allows households to make well-informed financial decisions. The methodology is based on the use of financial theories, data-driven modeling, normalization, and sensitivity analysis. The above framework helps bridge the gap between quantitative finance and personal financial planning.

***Index Terms*—Financial Fragility, Household Finance, Quantitative Finance, Risk Modeling, Portfolio Allocation, Recommendation Systems, Financial Resilience, Data Science**

I. Introduction

In the new financial order, an individual has the primary role of managing investments, savings, and financial planning. However, the majority of the population does not have access to structured analytical tools for assessing financial status before making investment decisions. This has led to many investment decisions based on short-term trends, social media influence, and general risk-tolerance surveys, which may not accurately represent an individual's financial capacity.

Financial fragility refers to the inability of an individual to withstand economic shocks such as income reduction, rising inflation, increases in interest rates, and stock market fluctuations. Financial fragility has been widely researched in health and economic systems, but it has not been sufficiently studied in the context of individual financial decisions.

This paper introduces a Household Financial Fragility Index (HFFI), which quantifies financial vulnerability across multiple dimensions of household finance. Unlike traditional investment frameworks that focus primarily on returns and volatility, this approach evaluates whether a household is structurally capable of sustaining financial risk.

The proposed system further integrates the fragility index with a quantitative strategy engine and a recommendation system. This ensures that investment decisions are aligned not only with market conditions but also with the household's financial resilience.

II. Literature Review

The concept of fragility was originally developed in clinical research, where Walsh *et al.* proposed the Fragility Index for evaluating the robustness of statistically significant results. Fragility is defined as the number of data points that must change for results to become insignificant [1].

Similarly, in the context of financial systems, researchers have analyzed fragility from a macroeconomic perspective. For example, Nicola Gennaioli, Andrei Shleifer, and Robert Vishny identified the role of leverage, behavioral biases, and mispricing in creating financial fragility [3]. Their work highlights the importance of fragility in financial systems and shows that it should not be overlooked in favor of expected returns alone.

From a household perspective, the work of Annamaria Lusardi, Daniel Schneider, and Peter Tufano emphasizes how emergency savings and the capacity to handle unexpected expenses contribute to household financial fragility [2]. Their research indicates that many households are not prepared for short-term financial shocks.

Moreover, portfolio theory developed by Harry Markowitz provides a basis for evaluating investment risks through diversification and volatility [4]. However, this theory does not consider households' underlying financial conditions.

Existing research does not provide a comprehensive framework that integrates household financial conditions, portfolio risks, and macroeconomic risks into one unified decision-making model. This paper addresses that gap by proposing a composite fragility index linked to investment strategy recommendations.

III. Methodology

The proposed methodology consists of three key components: fragility index construction, quantitative strategy mapping, and recommendation system design.

A. Fragility Index Construction

The Household Financial Fragility Index is defined as

$$\text{HFFI} = 100\Big[w_1 L_n + w_2 D_n + w_3 E_n + w_4 P_n + w_5 M_n + \lambda(L_n D_n)\Big] \quad (1)$$

where

$$\begin{aligned} L_n &: \text{ Liquidity fragility,} \\ D_n &: \text{ Debt fragility,} \\ E_n &: \text{ Expense fragility,} \\ P_n &: \text{ Portfolio fragility,} \\ M_n &: \text{ Macroeconomic fragility.} \end{aligned} \quad (2)$$

Each of these components is normalized in the range $[0, 1]$.

Liquidity fragility measures the adequacy of emergency savings relative to essential expenses. Debt fragility captures repayment burden and leverage using debt-to-income ratios. Expense fragility evaluates the rigidity of household spending. Volatility, concentration, and drawdown risks are covered by portfolio fragility. Inflation risk, interest-rate risk, and unemployment risks are covered by macroeconomic fragility.

An interaction term is included to capture nonlinear effects, where low liquidity combined with high debt significantly increases financial vulnerability.

B. Weight Selection and Validation

The weights assigned to each component are initially determined using theory-driven calibration. Liquidity and debt are given higher importance due to their direct relationship with financial distress. Portfolio risk is assigned moderate importance due to its relevance in investment decision-making, while expense rigidity and macroeconomic exposure are assigned moderate weights.

To ensure robustness, the model incorporates sensitivity analysis and can be extended using regression-based calibration or optimization techniques for empirical validation.

C. Quantitative Strategy Mapping

The computed HFFI score is used to determine appropriate investment strategies, as shown in Table I.

TABLE I
HFFI RANGE AND INVESTMENT STRATEGY

HFFI Range	Strategy
0–30	Growth
31–60	Balanced
61–80	Defensive
81–100	Capital Preservation

Each strategy corresponds to a predefined allocation across asset classes such as liquid savings, bonds, and equities. This ensures that investment decisions are aligned with the financial resilience of the household.

D. Recommendation System

A rule-based recommendation system generates personalized financial advice based on fragility components. For example, high liquidity fragility results in recommendations to increase emergency savings, while high debt fragility leads to suggestions for reducing leverage.

The recommendation system enhances interpretability by translating quantitative results into actionable insights.

E. Data and Implementation

The system uses household finance data, macroeconomic data, and portfolio risk metrics. The implementation process involves data processing and modeling techniques based on Python. The results are displayed in a terminal-like format.

IV. HFFI MODEL FORMULATION AND COMPONENT DESIGN

A. Fragility Index Construction

The final Household Financial Fragility Index is defined as:

$$\text{HFFI} = 100\Big[0.25L_n + 0.25D_n + 0.15E_n + 0.20P_n + 0.15M_n + 0.10(L_n D_n)\Big] \quad (3)$$

where L_n represents normalized liquidity fragility, D_n represents normalized debt fragility, E_n represents normalized expense fragility, P_n represents normalized portfolio fragility, and M_n represents normalized macroeconomic fragility. The term $0.10(L_n D_n)$ represents an interaction penalty for households with both low liquidity and high debt.

A household becomes financially fragile because of several connected risks. A single variable cannot fully explain fragility. For example, income alone does not show whether the household has savings, savings alone does not show debt burden, and portfolio risk alone does not show whether the household can survive an economic downturn. Therefore, the model uses five dimensions: liquidity, debt, expenses, portfolio exposure, and macroeconomic sensitivity.

1) Liquidity Fragility: Liquidity fragility measures the household's emergency survival capacity. It is computed as:

$$L = 1 - \min\left\{\frac{S_L}{6E_m}, 1\right\} \quad (4)$$

where S_L denotes liquid savings and E_m denotes monthly essential expenses. The six-month emergency buffer is used because it is a common personal finance benchmark for surviving income loss, medical emergencies, or economic downturns. If liquid savings are sufficient to cover six months of essential expenses, then liquidity fragility approaches zero. If the household has no liquid savings, then liquidity fragility approaches one. Thus, higher savings reduce fragility, while higher essential expenses increase fragility.

2) Debt Fragility: Debt fragility captures both short-term repayment pressure and long-term leverage burden. It is computed as:

$$D = 0.6\left(\frac{DP_m}{Y_m}\right) + 0.4\left(\frac{D_T}{Y_a}\right) \tag{5}$$

where DP_m is monthly debt payment, Y_m is monthly income, D_T is total debt, and Y_a is annual income. The monthly debt payment ratio receives a weight of 0.6 because debt stress is experienced through monthly cash flow obligations. The total debt-to-income ratio receives a weight of 0.4 because long-term leverage is also important, although it is less immediate than monthly repayment pressure. Therefore, higher debt burden increases fragility, while higher income capacity reduces fragility.

3) Expense Fragility: Expense fragility measures spending rigidity and is defined as:

$$E = \frac{E_m}{E_T} \tag{6}$$

where E_T denotes monthly total expenses. A household with a high share of essential expenses has less flexibility to reduce spending during financial stress. Therefore, as essential expenses increase relative to total expenses, expense fragility increases.

4) Portfolio Fragility: Portfolio fragility captures investment risk exposure and is defined as:

$$P = 0.4\sigma_p + 0.3C + 0.3DD \tag{7}$$

where σ_p is portfolio volatility, C is portfolio concentration, and DD is expected drawdown.

Portfolio fragility is not only measured through volatility. A household portfolio can be financially fragile for multiple reasons. Volatility measures how much the portfolio value fluctuates, concentration measures whether the portfolio is overly dependent on one stock, sector, asset, or theme, and expected drawdown measures the possible downside loss during adverse market conditions.

The weights 0.4, 0.3, and 0.3 are selected to balance the three major sources of portfolio risk. Volatility receives the highest weight of 0.4 because it is the broadest and most commonly used measure of portfolio risk. Concentration receives a weight of 0.3 because lack of diversification increases fragility. Expected drawdown also receives a weight of 0.3 because downside loss directly affects household stability.

The portfolio weights satisfy:

$$0.4 + 0.3 + 0.3 = 1 \tag{8}$$

This ensures that the portfolio fragility score remains balanced and interpretable.

The formula also behaves logically. If portfolio volatility increases, then P increases. If concentration increases, then P increases. If expected drawdown increases, then P also increases. Therefore,

$$\sigma_p \uparrow \Rightarrow P \uparrow, \quad C \uparrow \Rightarrow P \uparrow, \quad DD \uparrow \Rightarrow P \uparrow \tag{9}$$

Thus, higher investment risk leads to higher portfolio fragility.

5) Macroeconomic Fragility: Macroeconomic fragility captures exposure to external economic shocks and is defined as:

$$M = 0.4I + 0.3R + 0.3U \tag{10}$$

where I represents inflation sensitivity, R represents interest-rate sensitivity, and U represents unemployment sensitivity. Inflation receives the highest weight because it directly affects everyday expenses. Interest-rate and unemployment sensitivity are also included because they influence borrowing costs and income stability.

6) Normalization: All raw components are normalized to the interval $[0, 1]$ before being used in the final HFFI formula:

$$L_n, D_n, E_n, P_n, M_n \in [0, 1] \tag{11}$$

Normalization is necessary because each raw variable has a different scale. For a raw component X, the normalized value is computed as:

$$X_n = \begin{cases} 0, & X \le X_{\min}, \\ 1, & X \ge X_{\max}, \\ \dfrac{X - X_{\min}}{X_{\max} - X_{\min}}, & X_{\min} < X < X_{\max}. \end{cases} \tag{12}$$

This ensures that no component dominates the index only because of its numerical scale.

7) Weight Justification: The final weights are selected as:

$$\begin{aligned} w_1 &= 0.25, \qquad w_2 = 0.25, \qquad w_3 = 0.15, \\ w_4 &= 0.20, \qquad w_5 = 0.15, \qquad \lambda = 0.10. \end{aligned} \tag{13}$$

Liquidity receives a weight of 0.25 because emergency savings are the first line of defense against financial shocks. Debt also receives a weight of 0.25 because debt creates fixed obligations and reduces financial flexibility. Portfolio fragility receives a weight of 0.20 because this project connects household fragility with investment strategy. Expense rigidity and macroeconomic sensitivity each receive a weight of 0.15 because they are important but should not dominate liquidity and debt.

The interaction coefficient $\lambda = 0.10$ is used to capture nonlinear financial stress when liquidity fragility and debt fragility are both high. It amplifies risk but does not dominate the overall model.

8) Portfolio Allocation and Live Portfolio Monitoring: The proposed framework also evaluates household-level portfolio allocation structure. The objective is not only to compute financial fragility but also to understand how household assets are distributed across investment categories. The allocation model provides interpretable measurements of portfolio composition and enables dynamic recommendation generation.

The allocation framework separates household assets into three major components:

- Equity allocation
- Bond allocation
- Liquid savings allocation

Unlike traditional portfolio frameworks that treat cash as a generic portfolio component, the proposed system explicitly models liquid savings because household liquidity directly contributes to financial resilience. Liquid savings represent emergency reserves, savings accounts, checking balances, and immediately accessible household buffers. Therefore, liquid savings are modeled separately from market-invested assets and are used as a resilience indicator inside the HFFI framework.

a) Equity Allocation: Equity allocation measures the proportion of household financial assets invested in equity-based instruments such as stocks, equity ETFs, growth-oriented mutual funds, and retirement equity holdings. Equity exposure is important because it represents the household's participation in long-term market growth, but it also increases exposure to volatility and drawdown risk.

The equity allocation percentage is computed as:

$$A_E = \frac{V_E}{V_T} \times 100 \tag{14}$$

where A_E is the equity percentage, V_E is the current market value of all equity-based holdings, and V_T is the total household financial asset value considered in the portfolio allocation calculation. A higher equity percentage indicates a more growth-oriented portfolio, while a lower equity percentage indicates a more conservative portfolio.

b) Bond Allocation: Bond allocation measures the proportion of household financial assets invested in fixed-income instruments such as treasury securities, government bonds, corporate bonds, and bond exchange-traded funds. Bonds are included because they generally provide more stable returns than equities and can reduce portfolio volatility during periods of market stress.

The bond allocation percentage is computed as:

$$A_B = \frac{V_B}{V_T} \times 100 \tag{15}$$

where A_B is the bond percentage and V_B is the current value of all fixed-income holdings. A higher bond allocation generally improves portfolio stability and reduces sensitivity to equity-market volatility. Therefore, financially fragile households may require higher defensive bond exposure relative to financially stable households.

c) Liquid Savings Allocation: The framework uses liquid savings instead of generic cash allocation because liquidity directly contributes to household survivability during financial stress. Liquid savings include checking accounts, savings accounts, emergency reserves, and other immediately accessible funds. This distinction is important because liquid savings are not treated only as idle cash; rather, they represent the household's ability to absorb income shocks, medical expenses, inflationary pressure, and unexpected obligations.

The liquid savings allocation percentage is defined as:

$$A_L = \frac{S_L}{V_T} \times 100 \tag{16}$$

where A_L is the liquid savings percentage and S_L is the household's liquid savings. Higher liquid savings improve the household's ability to withstand short-term shocks and therefore reduce liquidity fragility. This formulation is more consistent with the HFFI framework than a generic cash allocation because emergency savings are a direct component of household resilience.

d) Portfolio Allocation Constraint: The allocation structure satisfies:

$$A_E + A_B + A_L = 100 \tag{17}$$

This ensures that the household's financial assets are fully represented across growth exposure, defensive exposure, and liquidity resilience.

e) Live Portfolio Monitoring: For live portfolio monitoring, the system calculates current market value and unrealized profit/loss separately for each asset. This allows the HFFI recommendation engine to reflect both household fragility and the current state of investment holdings.

The current market value of asset i is computed as:

$$CV_i = q_i \times p_i \tag{18}$$

where CV_i is current value, q_i is the number of units of asset i held by the household, and p_i is the latest live market price.

The unrealized profit or loss for asset i is computed as:

$$UPL_i = CV_i - I_i \tag{19}$$

The unrealized profit or loss percentage is defined as:

$$UPL_i^{\%} = \frac{UPL_i}{I_i} \times 100 \tag{20}$$

where UPL_i is unrealized profit or loss and I_i is the invested amount. These live portfolio metrics allow the system to monitor whether the household is experiencing gains, losses, or increasing concentration risk. As a result, the recommendation engine can update its outputs based on both static household features and changing market conditions.

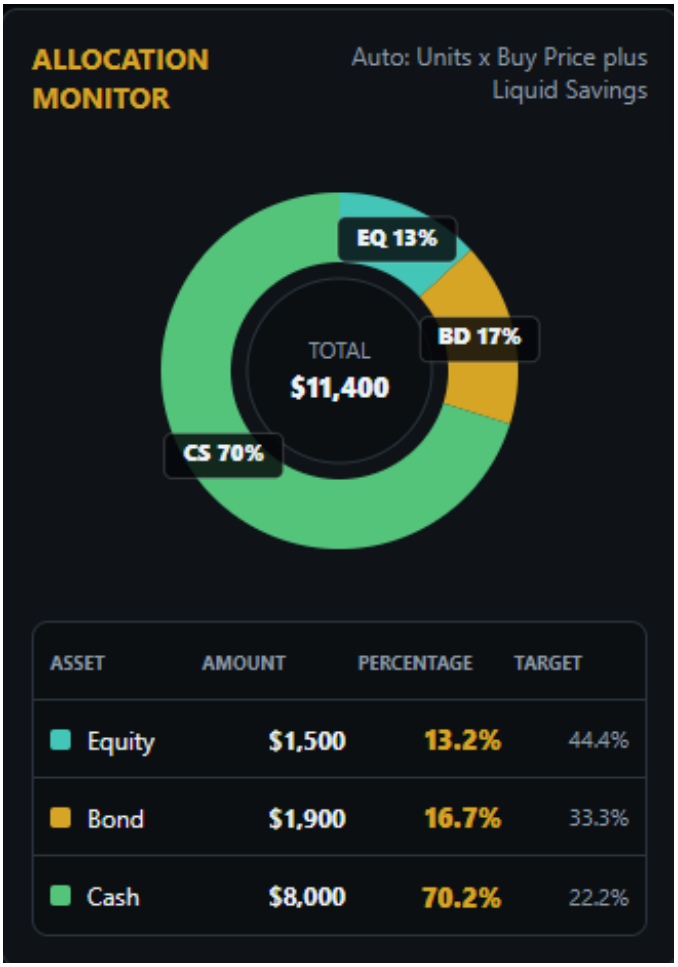

Fig. 1. Allocation monitor showing household exposure across equity, bond, and liquid savings components.

V. Mathematical Proof and Model Justification

A. Convex Combination Property

The selected component weights satisfy:

$$0.25 + 0.25 + 0.15 + 0.20 + 0.15 = 1 \tag{21}$$

Also, each weight is nonnegative:

$$w_i \geq 0, \quad i = 1, 2, 3, 4, 5. \tag{22}$$

Therefore, the linear part of the HFFI model is a convex combination of normalized fragility components. This ensures that the model remains interpretable, bounded, and balanced.

B. Boundedness Proof

Since all normalized components satisfy:

$$0 \leq L_n, D_n, E_n, P_n, M_n \leq 1 \tag{23}$$

and all weights are nonnegative, the weighted sum is bounded as:

$$0 \leq 0.25L_n + 0.25D_n + 0.15E_n + 0.20P_n + 0.15M_n \leq 1. \tag{24}$$

The interaction term is also bounded because:

$$0 \leq L_n D_n \leq 1. \tag{25}$$

Thus:

$$0 \leq 0.10(L_n D_n) \leq 0.10. \tag{26}$$

Therefore:

$$0 \leq \text{HFFI} \leq 100(1 + 0.10), \tag{27}$$

which gives:

$$0 \leq \text{HFFI} \leq 110. \tag{28}$$

For practical interpretability, the score is capped at 100.

C. Monotonicity Proof

The HFFI score must increase when any fragility component increases. For liquidity:

$$\frac{\partial \text{HFFI}}{\partial L_n} = 100(0.25 + 0.10D_n). \tag{29}$$

Since $D_n \geq 0$, this derivative is positive. For debt:

$$\frac{\partial \text{HFFI}}{\partial D_n} = 100(0.25 + 0.10L_n). \tag{30}$$

Since $L_n \geq 0$, this derivative is also positive. For the remaining components:

$$\frac{\partial \text{HFFI}}{\partial E_n} = 100(0.15) > 0, \tag{31}$$

$$\frac{\partial \text{HFFI}}{\partial P_n} = 100(0.20) > 0, \tag{32}$$

$$\frac{\partial \text{HFFI}}{\partial M_n} = 100(0.15) > 0. \tag{33}$$

Therefore, if any fragility component increases, the final HFFI score increases. This proves that the formula behaves logically.

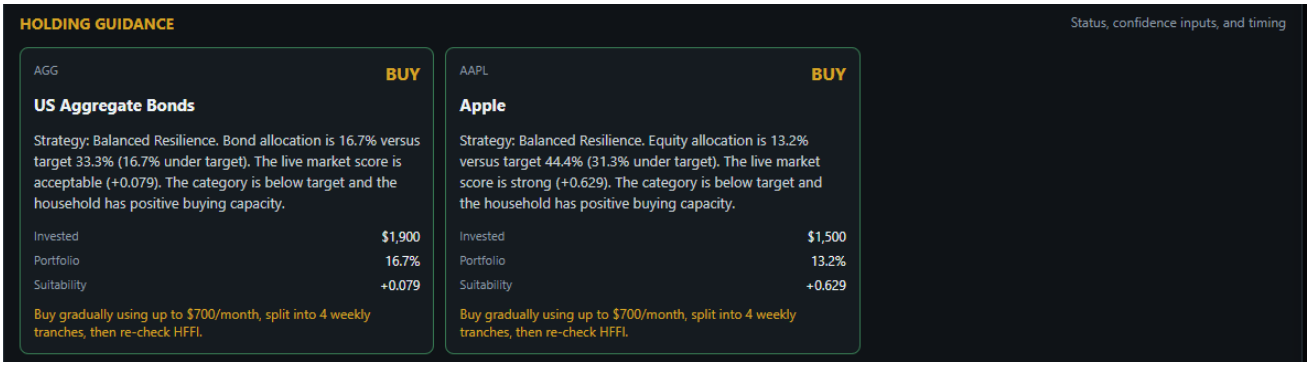

Fig. 2. Holding guidance panel showing asset-level recommendation status, allocation gap, and plain-language investment rationale.

D. Interaction Term Proof

The interaction term is:

$$0.10(L_n D_n). \tag{34}$$

If both liquidity fragility and debt fragility are high, then:

$$L_n \to 1, \quad D_n \to 1. \tag{35}$$

and therefore:

$$L_n D_n \to 1. \tag{36}$$

In this case, the interaction penalty approaches 0.10, which increases the final fragility score. If either liquidity fragility or debt fragility is low, then:

$$L_n D_n \to 0. \tag{37}$$

and the penalty becomes small. This proves that the interaction term activates strongly only when both liquidity risk and debt risk are high.

E. Calibration Explanation

The proposed values are selected using theory-driven calibration. This means that the weights are not randomly assigned. They are selected based on financial theory, household risk logic, interpretability, model balance, and sensitivity-testing requirements.

The selected weights are initial baseline values rather than universal constants. In future work, these weights can be estimated using regression analysis, machine learning, principal component analysis, optimization techniques, or empirical calibration on real household datasets.

VI. Results and Analysis

A. Experimental Setup

To evaluate the proposed Household Financial Fragility Index (HFFI), the system was implemented using Python and tested on simulated household financial profiles. The dataset consists of multiple hypothetical households with varying characteristics such as income levels, savings, debt burden, expense structure, and portfolio risk exposure.

The system pipeline includes:

- Data preprocessing and validation
- Feature engineering
- Fragility computation using the HFFI formula
- Strategy mapping based on fragility score
- Recommendation generation
- Stress testing under simulated economic shocks

Additionally, real-time macroeconomic data such as inflation, unemployment rate, and interest rates were integrated using external APIs to enhance the realism of the model.

B. Feature Engineering and Model Behavior

The implementation introduced additional derived variables to improve the robustness of the fragility model. Feature engineering is important because raw household variables do not always directly reveal financial risk. For example, total debt alone does not show whether repayment pressure is manageable unless it is compared with income. Similarly, liquid savings alone does not show resilience unless it is compared with essential monthly expenses.

The main engineered features include:

- Debt Service Ratio, computed as monthly debt payment divided by monthly income
- Debt-to-Income Ratio, computed as total debt divided by annual income
- Liquidity Buffer, computed as liquid savings relative to six months of essential expenses
- Portfolio Risk Interaction, computed from volatility and concentration effects
- Macro Stress Index, which summarizes inflation, interest-rate, and unemployment pressure

These engineered features allow the model to capture deeper financial relationships and improve sensitivity to household conditions. The debt-service ratio measures short-term cash-flow pressure. The debt-to-income ratio captures long-term leverage burden. The liquidity buffer measures emergency survivability. Portfolio interaction terms capture the combined effect of market volatility and lack of diversification. The

macro stress index links household fragility to broader economic conditions.

Fig. 3 shows the engineered feature behavior across the evaluated household profiles. The debt-to-income ratio plot illustrates how highly leveraged households tend to exhibit higher HFFI scores, although some extreme leverage cases appear as outliers. The liquidity-buffer plot shows an inverse relationship between emergency savings capacity and HFFI, indicating that stronger liquidity buffers reduce measured fragility. The debt-service ratio by risk band shows that repayment burden is concentrated among higher-fragility households, while the macro stress index panel indicates that external economic stress affects households across all fragility bands. These diagnostics support the interpretation that the HFFI framework responds to both household-level financial pressure and macroeconomic risk exposure.

C. Scenario-Based Evaluation

1) Low Fragility (Stable Household): Households with high income, strong savings, and low debt produced HFFI scores below 30.

Observations:

- Liquidity fragility was minimal
- Debt fragility remained low
- Portfolio risk had limited impact

Result:

- Strategy: Growth
- Recommendation: Higher allocation to equities

2) Moderate Fragility: Households with moderate savings, moderate debt, and rigid expenses produced HFFI scores between 30 and 60.

Observations:

- Expense rigidity increased fragility
- Debt contributed significantly
- Macroeconomic sensitivity influenced results

Result:

- Strategy: Balanced
- Recommendation: Diversified portfolio allocation

3) High Fragility: Households with low liquidity, high debt, and high portfolio risk produced HFFI scores above 60.

Observations:

- Liquidity fragility dominated
- Debt interaction term amplified risk
- Portfolio volatility increased vulnerability

Result:

- Strategy: Defensive
- Recommendation: Increase liquid savings and reduce risk exposure

D. Stress Testing Analysis

A stress scenario was introduced by simulating a 20% reduction in household income.

Key Findings:

- HFFI scores increased significantly across all households
- Many households shifted from moderate to high fragility
- Investment strategies shifted toward capital preservation

This demonstrates that the model effectively captures financial sensitivity under adverse economic conditions.

E. Recommendation System Evaluation

The recommendation system was enhanced to include:

- Portfolio suitability scoring
- Risk-adjusted allocation suggestions
- Financial action recommendations

Examples include:

- Increase emergency funds if liquidity fragility is high
- Reduce leverage if debt fragility is high
- Diversify investments if portfolio fragility is high

F. Recommendation Layer

The project extends the rule-based HFFI framework with a supervised machine learning recommendation layer. The recommendation engine combines household financial characteristics, macroeconomic indicators, portfolio structure, HFFI outputs, and market conditions to rank possible investment candidates. This design allows the system to move beyond static rules and generate recommendations that are both household-aware and market-aware.

The recommendation framework uses the following machine learning models:

1) Random Forest Classifier
2) Gradient Boosting Classifier
3) Ensemble probability score

The prediction target is defined as:

$$T_{j,t+1} = \begin{cases} 1, & r_{j,t+1} > 0, \\ 0, & r_{j,t+1} \leq 0, \end{cases} \tag{38}$$

where $T_{j,t+1}$ is the next-month direction label for market candidate j and $r_{j,t+1}$ is its next-month return. The recommendation layer uses multiple feature groups to improve predictive robustness.

a) Household Features: Household features capture the financial condition of the household. These include monthly income, essential expenses, liquid savings, total debt, monthly debt payment, HFFI score, debt service ratio, debt-to-income ratio, and liquidity buffer. These variables ensure that recommendations are not based only on expected market return, but also on the household's ability to absorb investment risk.

b) Macro Features: Macroeconomic features capture the external financial environment. These include inflation, unemployment rate, federal funds rate, and macro stress index. These variables allow the model to adjust recommendations when broader economic conditions become more unstable.

c) Portfolio Features: Portfolio features describe the household's current investment structure. These include equity percentage, bond percentage, liquid savings percentage, portfolio volatility, expected drawdown, and portfolio concentration. These variables allow the model to identify whether the household is overexposed to risky assets or underallocated to defensive assets.

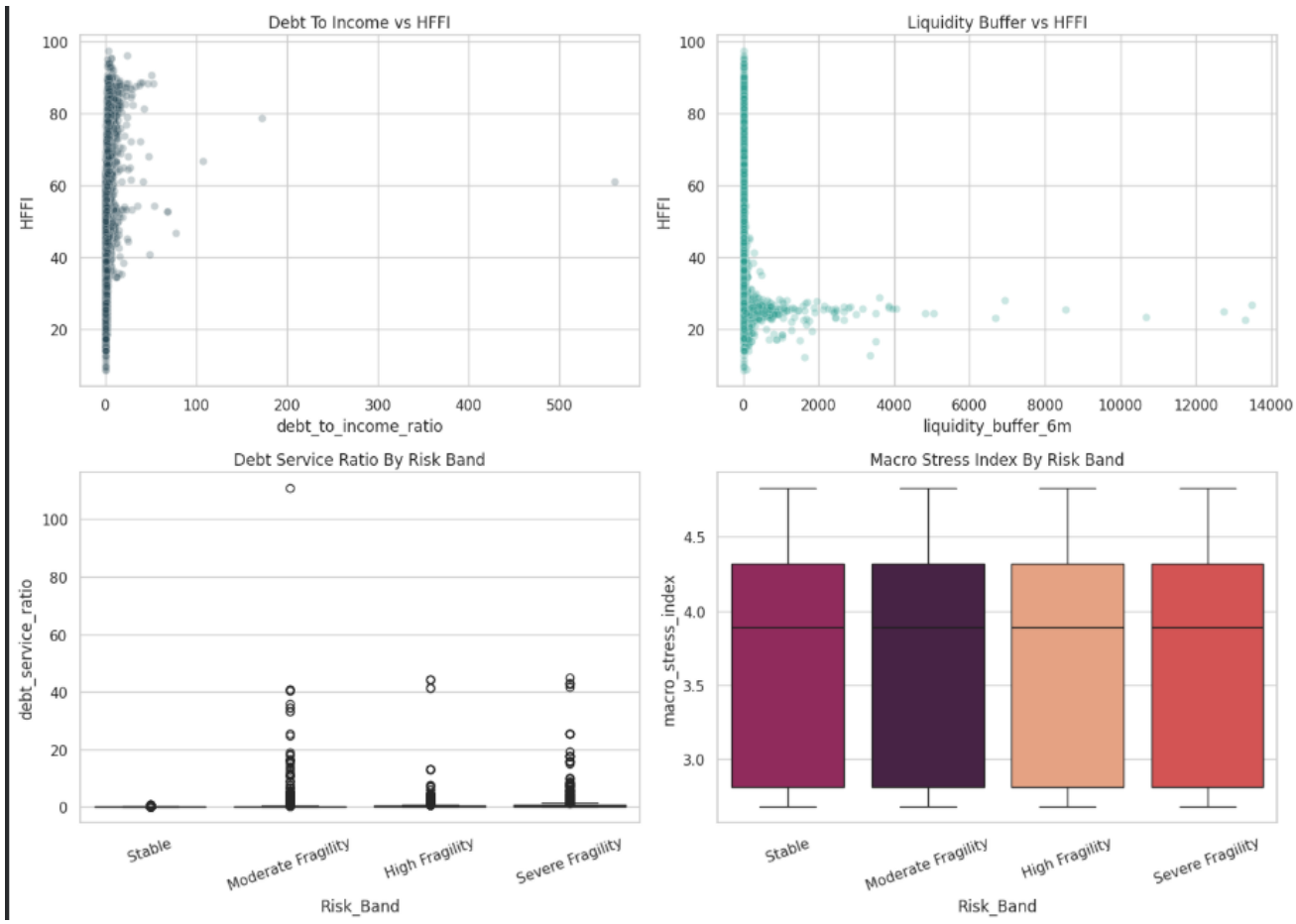

Fig. 3. Feature engineering diagnostics showing relationships between debt-to-income ratio, liquidity buffer, debt-service ratio, macro stress index, and HFFI risk bands.

d) Market Features: Market features describe current and historical behavior of investment candidates. These include market return, market volatility, market drawdown, momentum score, and safety score. These variables help the model distinguish between attractive growth opportunities and assets that may expose fragile households to excessive downside risk.

The ensemble machine learning probability is computed as:

$$\hat{p}_{ML} = \frac{\hat{p}_{RF} + \hat{p}_{GB}}{2} \tag{39}$$

where $\hat{p}_{RF}$ is the Random Forest probability and $\hat{p}_{GB}$ is the Gradient Boosting probability. Using an ensemble probability reduces dependence on a single model and improves stability in recommendation scoring.

The Random Forest and Gradient Boosting outputs in Fig. 4 and Fig. 5 show that both supervised models produce comparable predictive behavior. The measured accuracy is approximately 42.3%, which indicates that the model is not intended to be interpreted as a perfect market-timing classifier. Instead, it is used as one component inside the recommendation score. The class-level metrics show stronger precision for positive-return months than for negative-return months, while the recall pattern indicates that the models remain conservative and imperfect under noisy market conditions. This supports the use of an ensemble probability rather than relying on a single classifier output.

```
Gradient Boosting Accuracy: 0.4227921326648669

Gradient Boosting Classification Report:

              precision    recall  f1-score   support

           0       0.32      0.67      0.44     86655
           1       0.64      0.30      0.41    172645

    accuracy                           0.42    259300
   macro avg       0.48      0.48      0.42    259300
weighted avg       0.54      0.42      0.42    259300


Gradient Boosting Confusion Matrix:

[[ 57770  28885]
 [120785  51860]]
```

Fig. 4. Random Forest classification report and confusion matrix for next-month market direction prediction.

```
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Random Forest Confusion Matrix:

[[ 57770  28885]
 [120785  51860]]
```

Fig. 5. Gradient Boosting classification report and confusion matrix for next-month market direction prediction.

G. Data Science Recommendation Score

Each asset is ranked using a combined Data Science recommendation score. The score combines predictive market probability with household financial capacity, allocation need, and downside protection.

The recommendation score is defined as:

$$S_{DS} = 0.35\hat{p}_{ML} + 0.25C_H + 0.20S_A + 0.20S_D \tag{40}$$

The HFFI capacity component is computed as:

$$C_H = \frac{100 - \text{HFFI}}{100} \tag{41}$$

where C_H is HFFI capacity. This term converts financial fragility into risk capacity. A lower HFFI score produces a higher capacity value, indicating that the household can tolerate more investment risk. A higher HFFI score produces a lower capacity value, indicating that the household requires more cautious recommendations.

The allocation gap is defined as:

$$G_A = A_T - A_C \tag{42}$$

The allocation score is computed as:

$$S_A = \text{clip}(0.5 + G_A, 0, 1) \tag{43}$$

where G_A is the allocation gap, A_T is target allocation, A_C is current allocation, and S_A is the allocation score. The allocation score rewards assets or asset classes that help move the household toward the recommended allocation structure. The clipping operation prevents the score from falling below 0 or exceeding 1.

The downside protection score is defined as:

$$S_D = 1 - |\sigma_m + DD_m| \tag{44}$$

where S_D is downside protection, σ_m is market volatility, and DD_m is market drawdown. This term penalizes assets with high volatility and severe drawdown behavior. Therefore, assets with better downside protection receive higher recommendation scores, especially for financially fragile households.

The recommendation decision rule is:

$$\text{Action} = \begin{cases} \text{BUY Candidate}, & S_{DS} \geq 0.68, \\ \text{HOLD / WATCH}, & 0.42 \leq S_{DS} < 0.68, \\ \text{AVOID / REDUCE}, & S_{DS} < 0.42. \end{cases} \tag{45}$$

This scoring method ensures that market attractiveness alone does not dominate the recommendation. Household fragility, allocation gap, and downside risk are also included. As a result, the system can recommend higher-growth assets to financially stable households while guiding financially fragile households toward more resilient alternatives.

H. System Implementation and Visualization

The system was implemented as an interactive terminal-style application with a web-based frontend. Users can input financial details and receive the HFFI score, risk classification, investment strategy, portfolio recommendation, recommendation-layer explanation, and graphical visualization of fragility components, allocation, and stress comparison.

Fig. 6 shows the implemented web-based HFFI dashboard. The interface accepts household-level inputs such as monthly income, expenses, debt, savings, portfolio volatility, employment type, mortgage type, and dependents. Based on these inputs, the system computes the base-case HFFI score and evaluates the impact of a 20% income shock.

The dashboard output shows that for the sample household, the base-case HFFI score is 63.41, placing the household in the high-fragility category. Under a 20% income shock, the HFFI score increases to 64.77, confirming that the household remains highly fragile under adverse conditions. The system recommends a defensive strategy with higher liquid savings, higher bond exposure, and lower equity exposure.

Fig. 7 presents the overall distribution of HFFI scores across the dataset. The distribution shows that households are spread across multiple fragility levels, with a strong concentration in the moderate fragility range. This supports the claim that financial fragility is not uniform and varies significantly across households.

Fig. 8 shows the count of households within each risk band. The majority of households fall into the moderate-fragility category, followed by stable households, high-fragility households, and severe-fragility households. This confirms that the proposed index can classify households into meaningful financial-risk groups.

Fig. 9 shows the trend of mean and median HFFI scores by survey year. The results indicate that household fragility changes over time and can be influenced by broader economic

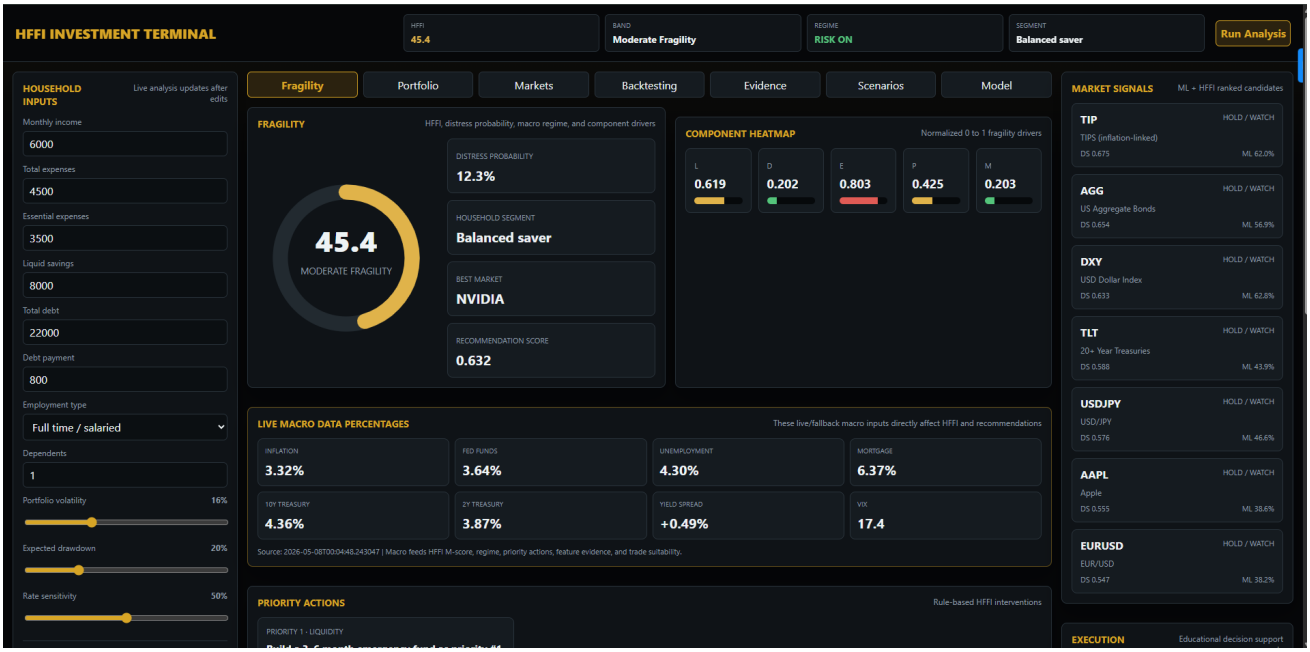

Fig. 6. Web-based implementation of the Household Financial Fragility Index dashboard.

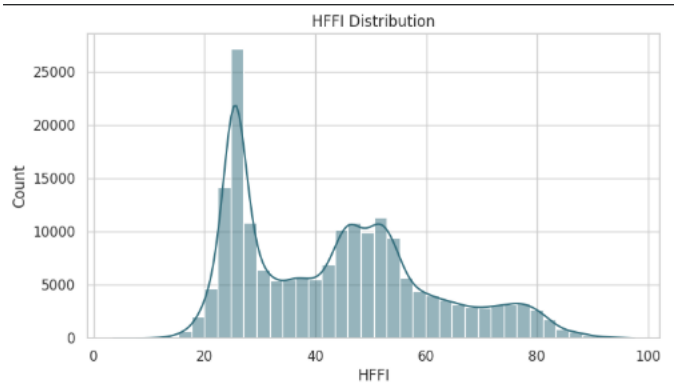

Fig. 7. Distribution of HFFI scores across simulated household profiles.

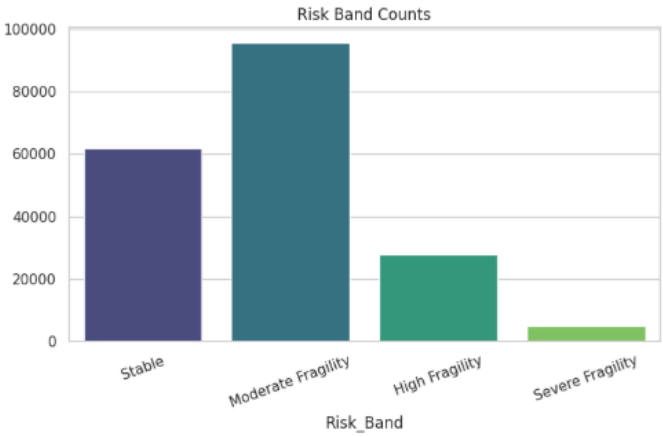

Fig. 8. Risk-band classification based on computed HFFI scores.

conditions. The declining pattern in later years suggests improvement in household financial stability for the evaluated dataset.

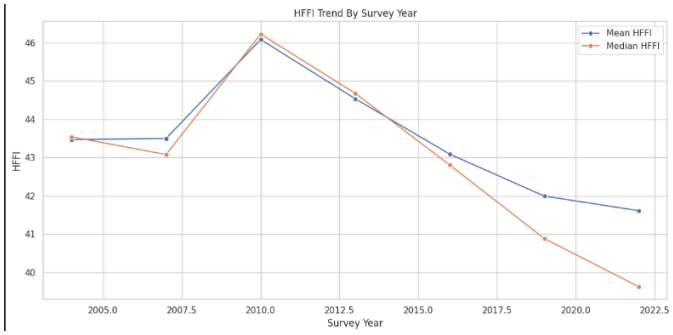

Fig. 9. Mean and median HFFI trend by survey year.

Fig. 10 shows the average realized next-month return by the top recommended market. The result indicates that recommended market categories produced different realized returns, demonstrating the importance of linking household fragility scores with investment strategy selection.

Overall, the visual results demonstrate the practical applicability of the proposed HFFI framework. The system not only computes a fragility score but also provides interpretable outputs, risk-band classification, stress-test comparison, strategy allocation, and recommendation explanations.

I. Model Validation Report Summary

The implementation was also evaluated through a structured validation report that tested reproducibility, end-to-end pipeline execution, boundary conditions, stress-test monotonicity, error behavior by risk band, distress probability

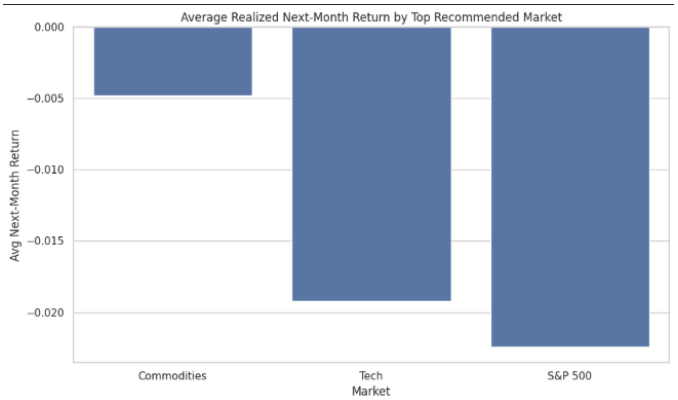

Fig. 10. Average realized next-month return by top recommended market.

calibration, range validity, and year-by-year score stability. The purpose of this validation layer is to ensure that the HFFI system behaves consistently before it is used for recommendation generation.

As shown in Fig. 11, the reproducibility test passed with a high correlation between repeated runs and a strong risk-band match rate. The end-to-end pipeline also passed, confirming that all required models were trained and executed. Boundary and edge cases produced valid bounded scores, and the stress-test pipeline passed because simulated adverse conditions increased HFFI monotonically. The validation report also identified caveats in severe-band error behavior and warnings in distress probability calibration and range monotonicity. These warnings are important because they show that the system is transparent about known limitations rather than hiding calibration weaknesses. Overall, the validation report supports the system's use as a decision-support framework while identifying areas for future calibration improvement.

J. Key Insights

- Financial instability is a multidimensional concept and cannot be measured by one single measure.
- Liquidity and debt have the highest influence on the HFFI index.
- Engineered features improve the interpretability of debt pressure, liquidity adequacy, portfolio exposure, and macroeconomic sensitivity.
- The HFFI distribution shows that most households fall into the moderate-fragility category.
- Stress testing confirms that income shocks increase household fragility scores.
- The dashboard implementation demonstrates that the HFFI framework can be applied as a practical decision-support system.
- Macro conditions significantly affect household risk.
- Interaction effects amplify financial stress in real-world scenarios.
- Explainable recommendations improve usability and decision-making clarity.

VII. Backtesting and Signal Verification

The recommendation system is evaluated using historical price data. The purpose of the backtesting process is to verify whether HFFI-based recommendations can produce more suitable household-level investment decisions than simple passive strategies. The backtest does not only measure return; it also evaluates risk, drawdown, turnover, and signal consistency.

The backtest compares three strategies:

1) HFFI Recommendation Strategy
2) Buy-and-Hold Strategy
3) Benchmark Strategy, such as SPY

The HFFI recommendation strategy uses the household fragility score, market features, portfolio allocation gap, and downside protection score to determine which asset should be recommended. The buy-and-hold strategy assumes that the investor maintains a fixed position without responding to changing HFFI or market signals. The benchmark strategy provides a market reference point, such as SPY, for comparison against a broad equity-market proxy.

The system rebalances the recommendation portfolio at a selected frequency and executes trades on the next trading day. This execution rule reduces look-ahead bias because the model does not assume that trades can be placed using information that would not have been available at the decision time.

The following metrics are used:

- Total Return
- Compound Annual Growth Rate
- Annualized Volatility
- Maximum Drawdown
- Sharpe Ratio
- Turnover
- Number of Trades

Total return measures the overall investment gain or loss over the evaluation period. Compound annual growth rate measures the annualized return after compounding. Annualized volatility measures return variability. Maximum drawdown measures the largest peak-to-trough decline and is especially important for financially fragile households. The Sharpe ratio compares return against volatility. Turnover and number of trades measure how frequently the strategy changes positions.

The system also includes signal verification. The latest backtest BUY, SELL, or HOLD action is compared with the current live recommendation engine. If both signals match, the recommendation is considered directionally consistent. If they differ, the system flags the result for review because market conditions, live prices, household inputs, and macroeconomic conditions may have changed.

This verification layer improves model transparency and helps prevent blind reliance on historical backtest outputs when live conditions no longer match the tested environment.

VIII. Monte Carlo Stress Testing and Counterfactual Analysis

To evaluate uncertainty, the system performs Monte Carlo stress testing by simulating multiple combinations of household and macroeconomic shocks. Instead of relying on a single

Test	Result	Verdict
T1: Score reproducibility (n=5,000)	r=0.9430, MAE=2.81, band match 90.2%	PASS
T2: End-to-end pipeline (n=200)	200/200, mean diff −6.41 pts	PASS with bias
T3: Boundary / edge cases	5/5 produce valid, bounded scores	PASS
T4: Stress test pipeline	7/7 scenarios run, monotonic ordering	PASS
T5: Error by band	Stable: MAE=0; Severe: MAE=12.0 (D/M clipping)	PASS with caveat
T6: Distress prob calibration	Anchor points off by up to 11.8%	WARN — known placeholder
T7: Range / monotonicity	D > 1.0 in 17.5% rows; M > 1.0 in 1.2%	WARN — clipping divergence
T8: Year-by-year stability	87.8% – 91.4% band match across all 7 years	PASS

Fig. 11. Validation report summary for HFFI reproducibility, stress behavior, calibration, range validity, and year-by-year stability.

deterministic stress scenario, the Monte Carlo framework generates a distribution of possible HFFI outcomes.

The simulation engine includes combinations of:

- Income shocks
- Inflation shocks
- Rate shocks
- Unemployment shocks
- Equity drawdowns

Income shocks test whether the household can remain stable if earnings decline. Inflation shocks test whether rising prices increase essential expenses. Rate shocks test whether borrowing costs and debt payments become more burdensome. Unemployment shocks test income instability, and equity drawdowns test whether portfolio losses increase financial fragility.

The output includes:

- Mean simulated HFFI
- Standard deviation
- 5th percentile HFFI
- Median HFFI
- 95th percentile HFFI
- Probability of severe fragility

The mean simulated HFFI summarizes the expected fragility level under uncertainty. The standard deviation measures dispersion across scenarios. The 5th, median, and 95th percentile values describe the range of possible outcomes. The probability of severe fragility estimates the likelihood that the household moves into the highest-risk category under simulated shocks.

The system also performs counterfactual analysis to estimate which household action would reduce HFFI the most. Examples include increasing liquid savings, paying down debt, reducing essential expenses, shifting equity exposure toward bonds or liquid savings, and diversifying concentrated portfolio risk.

Counterfactual analysis improves interpretability because it does not simply report that a household is fragile. Instead, it identifies which action would most effectively reduce fragility. This converts the HFFI framework from a diagnostic model into an actionable financial decision-support system.

IX. Conclusion

This paper presents a novel framework that integrates financial fragility measurement with quantitative investment strategy design. Using financial theory, data science, and explainable modeling, the system provides a structured and interpretable approach to household financial decision-making.

The proposed HFFI model successfully incorporates multiple dimensions of financial risk, including liquidity, debt, expenses, portfolio exposure, and macroeconomic conditions. The combination of strategy engine and recommendation system guarantees that all investment choices correspond to the degree of financial stability in the household and not only market dynamics.

In addition, the application of real-time data analysis, stress testing, supervised recommendation scoring, and signal verification improves the practical aspect of the developed system. As demonstrated by the results obtained, the proposed framework is able to successfully identify differences in financial stability and generate correct investment strategy.

On the whole, the described system successfully combines principles of quantitative finance and personal financial management.

X. Future Work

While the proposed framework provides a strong foundation for financial fragility analysis, several extensions can further enhance its capabilities:

1) **Optimization Using Machine Learning:** Further research may be carried out to develop the ability of the HFFI model to learn the best weights through regression analysis and machine learning.
2) **Advanced NLP Integration:** The current rule-based explanation system can be extended using advanced NLP models such as transformer-based architectures to generate more personalized and context-aware financial insights.
3) **Use of Real-Time Information on Financial Markets:** Inclusion of real-time prices of assets and stock market indexes can lead to improvement in accuracy of predictions.

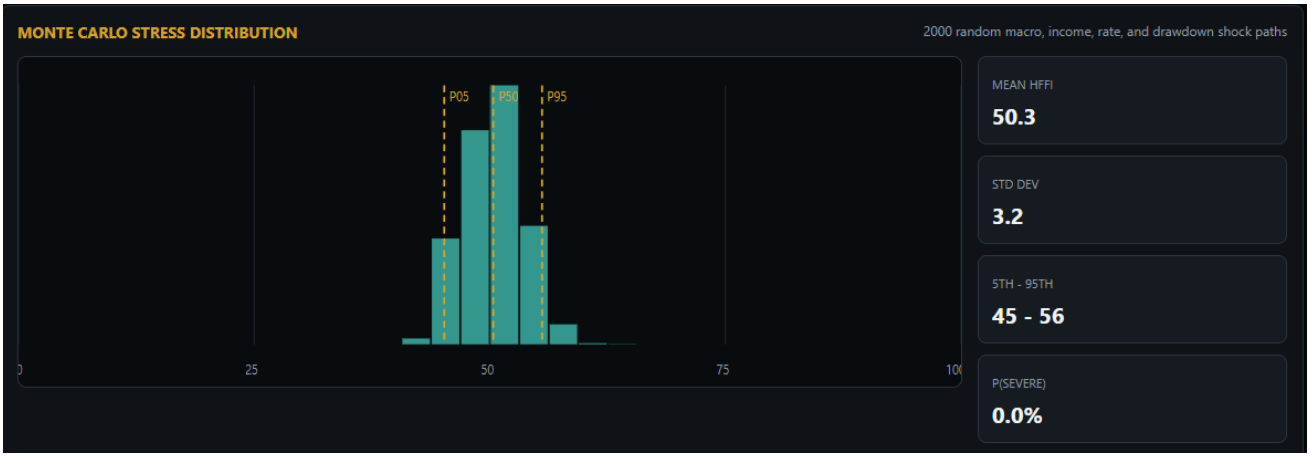

Fig. 12. Monte Carlo stress distribution showing simulated HFFI outcomes under randomized macroeconomic, income, interest-rate, and market-drawdown shock paths.

Fig. 13. Equity drawdown comparison showing peak-to-trough losses across the HFFI recommendation strategy, buy-and-hold strategy, and benchmark strategy.

4) **Global Economic Adaptability:** The current framework can be extended to work for multiple countries by considering specific region-related variables.
5) **Behavioral Finance Integration:** Future research can incorporate psychological and behavioral factors such as risk tolerance, spending behavior, and decision biases to improve the realism of the model.
6) **Automated Portfolio Optimization:** The recommendation system can be enhanced to include dynamic portfolio optimization using quantitative finance techniques.

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